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IntroStyle: Training-Free Introspective Style Attribution using Diffusion Features

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Abstract

Text-to-image (T2I) models have recently gained widespread adoption. This has spurred concerns about safeguarding intellectual property rights and an increasing demand for mechanisms that prevent the generation of specific artistic styles. Existing methods for style extraction typically necessitate the collection of custom datasets and the training of specialized models. This, however, is resource-intensive, time-consuming, and often impractical for real-time applications. We present a novel, training-free framework to solve the style attribution problem, using the features produced by a diffusion model alone, without any external modules or retraining. This is denoted as Introspective Style attribution (IntroStyle) and is shown to have superior performance to state-of-the-art models for style attribution. We also introduce a synthetic dataset of Artistic Style Split (ArtSplit) to isolate artistic style and evaluate fine-grained style attribution performance. Our experimental results on WikiArt and DomainNet datasets show that IntroStyle is robust to the dynamic nature of artistic styles, outperforming existing methods by a wide margin. Website: https://anandk27.github.io/IntroStyle

1. Introduction

Diffusion models have significantly advanced image synthesis by applying iterative denoising processes guided by input prompts. Models like Stable Diffusion [31], DALL-E 3 [1], and Imagen [33] have emerged as a powerful paradigm for T2I synthesis, demonstrating remarkable capabilities in generating high-quality images from textual descriptions. The success of diffusion-based approaches has led to their application in various domains, including layout-to-image generation, text-guided image generation, and even video synthesis. However, the widespread adoption and impressive performance of T2I modelss have also raised concerns about problems like copyright infringement. For effective performance, the models require large-scale pre-training on diverse datasets [35]. Since these datasets are collected automatically and dominantly from the web, it is difficult to control image provenance and avoid the collection of copyrighted imagery. This problem is compounded by the tendency of diffusion models to replicate elements from their training data [2]. The legal implications of training models on copyrighted images have become a subject of recent debate and litigation, with artists arguing that the unauthorized use of their works for AI training [6, 9, 11] constitutes copyright infringement [15, 30].

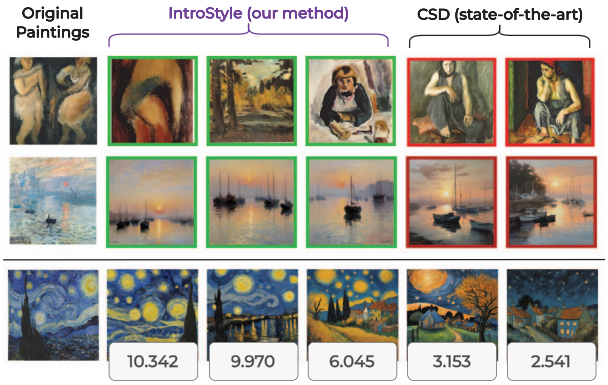

Figure 1. Introspective Style Attribution (IntroStyle). Top two rows: comparison of style attribution by IntroStyle and a SOTA method (CSD). A query image, the top-3 retrieval results of IntroStyle, and the top-2 retrieval results of CSD are shown from left to right. Green colors indicate correct and red incorrect retrievals. First row: WikiArt [34] dataset. Second row: proposed synthetic Artistic Style Split (ArtSplit) dataset. While the SOTA style attributions are biased by image semantics, failing to retrieve images on the style of the query, IntroStyle retrieves images of the correct style. Bottom: retrieval scores of IntroStyle for ArtSplit synthetic images generated to have the semantic of the query ("Starry Night") but different styles, demonstrating the effectiveness of IntroStyle as a metric for style measurement.

Current mitigation approaches include "unlearning" techniques to remove specific styles from AI models [44], though these require costly model retraining and may not fully address indirect style replication through alternative prompts [25]. Style "cloaking" methods [36] offer artists some protection but compromise the authentic viewing ex-

perience and place the burden of protection on creators. An alternative solution is to rely on attribution techniques [38, 43, 44], which can be used to attribute synthesized images to different artistic styles, allowing the post-hoc assessment of how much they might, for example, replicate an artist's artwork. These methods are also more practical since they simply return examples from the dataset used to train the model, ranked by similarity to the synthesized image. The main difficulty is to develop similarity functions that focus on style and are not confused by other image properties, such as similar image content or semantics.

To solve this problem, prior attribution approaches have proposed retraining the diffusion model [44] or adding external models to evaluate data similarity [38, 43]. This can add significant complexity since effective similarity models, like CLIP [27] or DINO [3], can be comparable in size to the diffusion model. On the other hand, prior diffusion model research has shown that different diffusion model layers develop distinct representations of higher and lower level image properties, such as structure vs color [42], producing features that can be leveraged to solve problems like image correspondence [40] or segmentation [41] without retraining. This led us to hypothesize that it may be possible to solve the attribution problem using the features produced by a diffusion model alone, without any external modules or retraining. We refer to this as *Introspective Style attribution* (`IntroStyle`).

In this work, we show that the hypothesis holds. We propose a simple and effective approach to differentiate image styles by analyzing the similarity of the features produced by a diffusion model for different images. We frame the denoising network of the diffusion model as an autoencoder whose encoder produces stylistic features. We then propose a metric to compare these features. While quite simple, this is shown to be quite effective for style retrieval. In fact, a comparison with the current state-of-the-art on popular style retrieval datasets shows that `IntroStyle` significantly outperforms prior models trained in the existing evaluation datasets, as illustrated in the top rows of Fig. 1. Although fine-tuned for style similarity, existing models still tend to retrieve images of similar semantics, whereas `IntroStyle` focuses much more uniquely on style.

Beyond models, the problem of style attribution suffers from a shortage of evaluation datasets. Creating such datasets is far from trivial since even the concept of "style" in art remains debatable. Nevertheless, many recognized artistic styles, such as surrealism and impressionism, are often associated with specific artists or movements. We use this social construct to define style as the collective global characteristics of an image identifiable with a particular artist or artistic movement. These characteristics encompass various elements such as color usage, brushstroke techniques, composition, and perspective. Under this definition, a single artist can produce art in several styles. Current datasets, primarily focusing on comparisons between authentic artworks, provide limited ability for evaluating how strongly style attribution methods disentangles style from semantics. From this point of view, an ideal dataset for evaluating style attributions should contain images with all pairs of styles and semantics, a requirement only attainable by synthetic data. While CSD [38] introduces synthetic data for generated image evaluation and GDA [43] uses custom diffusion to generate training datasets, these datasets lack fine-grained separation of artistic style and image semantics, which are heavily correlated in their images. This makes the resulting models poorly suited for enforcing copyright protections [17].

To address these problems, we introduce the *Artistic Style Split* (`ArtSplit`) dataset, inspired by [38]. This dataset combines real artistic image queries with synthetic retrieval images generated via diffusion models using carefully crafted prompts. For each authentic painting, we create two distinct prompts: a "semantic" prompt that excludes stylistic elements and a "style" prompt that omits semantic content. These prompts are fed into a diffusion model to generate corresponding images, enabling comprehensive coverage of the style $\times$ semantics space encompassing all combinations in real artistic queries. By analyzing the similarity between features extracted from query and synthetic images, we can precisely evaluate how feature representations balance style attribution against semantic similarity. Hence, style retrieval experiments on this dataset provide clear insights into how different models disentangle style from semantics. This is illustrated in the last row of Fig. 1, where the painting *Starry Night* by Vincent Van Gogh is used to retrieve images of paintings of the *Starry Night* semantic under various styles, using `IntroStyle`. Note how images of style closer to the reference have higher similarity scores, providing evidence that the `IntroStyle` representation favors style over semantics. This can be leveraged to implement style-based rejection sampling, preventing diffusion models from generating images of a specific style.

Overall, this paper makes the following contributions:

1. We formulate the problem of introspective style attribution without model retraining or external modules.
2. We propose `IntroStyle`, a method to generate style descriptors from images and a metric to compare image pairs according to style.
3. We introduce the `ArtSplit` dataset to isolate artistic style and evaluate fine-grained style attribution performance.
4. We perform extensive experiments showing that `IntroStyle` outperforms existing approaches of much higher complexity.

2. Related Work

2.1. Style Transfer

Style attribution and retrieval are related to style transfer, where the goal is to render one image in the style of another. While initially formulated using Markov random fields [48], style transfer raised in popularity with the rise of deep learning and the work of [8], which introduced Gram Matrices as style descriptors and an optimization framework for style transfer. Many variants have been proposed, including various regularizations to prevent distortions in reconstructed images [19] or representations of correlation beyond Gram matrices [4]. However, recent attribution work shows that Gram-style representations cannot guarantee effective style attribution [38].

Our work is inspired by more recent advances in style transfer using Generative Adversarial Networks (GANs), where methods like Adaptive Instance Normalization (AdaIN), which aligns the mean and variance of content features with those of style features, have been shown beneficial for real-time style transfer [12]. These ideas are central to the StyleGAN [13], where AdaIN layers are embedded into the generator architecture. This allows fine-grained control over different aspects of the generated images, from coarse features like pose to fine details like color schemes. More recently, diffusion-based style transfer has been implemented through AdaIN layer manipulation of initial timestep latent [5], and DDIM Inversion-based style latent reuse [20]. These approaches informed our investigation into diffusion model latent distributions.

2.2. Style-aware Text-to-Image Models

Early work in style-aware text-to-image generation built upon the success of neural style transfer techniques. Reed et al. [29] introduced the first GAN-based text-to-image model, while [47] improved image quality and resolution through a stacked architecture. More recently, diffusion models like DALL-E 2 [28] and Stable Diffusion [31] have demonstrated impressive capabilities in generating stylized images from text prompts. Several approaches have focused on explicitly representing and manipulating style in text-to-image models, such as style-based GAN architectures with fine-grained style control [18] or textual inversion methods [7, 14, 20] to learn new concepts and styles from a few examples, enabling personalized text-to-image generation. All these works require some fine-tuning of the generative model, whereas we pursue training-free solutions to the style attribution problem.

2.3. Data Attribution

While there is extensive literature on image retrieval, most methods focus on semantic similarity, i.e., the retrieval of images of similar content in terms of objects, scenes, etc. Style is a subtle property not well captured by representations, such as DINO [3] or CLIP [27], developed for this purpose. Few techniques have been developed specifically for style retrieval or attribution. Lee et al. [16] employed two separate neural network modules to facilitate style retrieval for image style and content. Recent works [32, 38, 43] instead train attribution models using contrastive objectives based on either Behance Artistic Media (BAM) dataset [32], a subset LAION Aesthetic Dataset [35] or synthetic style pairs. Ruta et al. [32] proposes a dual encoder to separate content and style with style features obtained using AdaIN layers, whereas CSD [38] and GDA [43] fine-tune a pre-trained CLIP or DINO backbone with a linear layer on top with the former training the entire network and latter training the linear layer alone. It is also possible to formulate data attribution as the unlearning of target images [44], but this requires a different model for each target. Unlike these methods, we seek an introspective solution to style attribution that does not require external models or attribution-specific training.

2.4. Diffusion Features

In the diffusion model literature, various works have studied the representations learned for denoising, showing that different layers of the network learn features that capture different image properties, like structure, color, or texture [22, 37, 42, 49]. Denoising Diffusion Autoencoders (DDAE) [45] leverage intermediate activations from pre-trained diffusion models for downstream tasks, demonstrating effectiveness in image classification through linear probing and fine-tuning. Recent advances include REPresentation Alignment (REPA) [46], which enhances semantic feature learning by aligning diffusion transformer representations with self-supervised visual features. Additionally, diffusion model features have shown promise in zero-shot applications like correspondence [40], segmentation [41] and classification [21]. These observations have inspired our hypothesis that style attribution can be solved introspectively.

3. Method

3.1. Diffusion

Diffusion models [39] are probabilistic models that define a Markov chain from which images are sampled by gradually denoising a normally distributed seed, using a neural network ϵ_θ. T2I models condition this sampling on a text prompt. A sentence describing the picture is encoded into a text embedding c by a text encoder and then used to condition the denoising network, usually via cross-attention. Stable Diffusion [31], a prominent T2I model, learns the chain in a latent space of a pre-trained image autoencoder. An encoder $\mathcal{E}$ first maps an image into a latent code z_0. In a

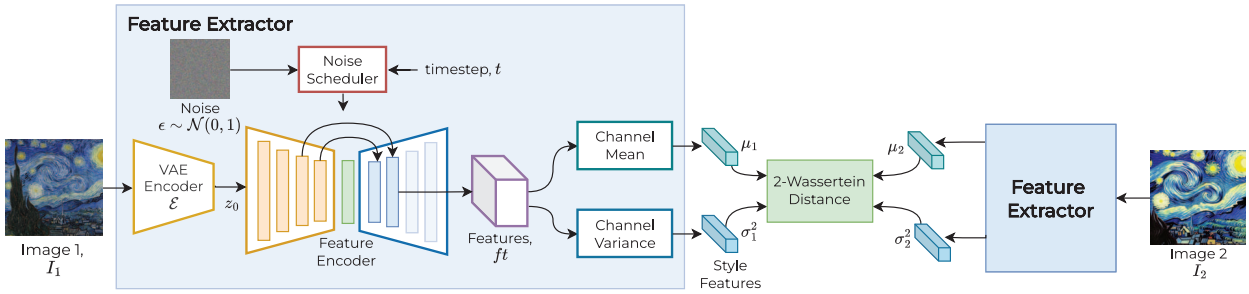

Figure 2. `IntroStyle` computation of style similarity: channel-wise mean μ and variance σ^2 are computed for the identified style layer features. Then 2-Wasserstein Distance is used to measure style similarities between a pair of images.

forward process, a sequence of noisier codes z_t are obtained by adding Gaussian noise to z_0 according to

$$z_t = \sqrt{\bar{\alpha}_t} z_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon_t, \quad \epsilon_t \sim N(0, I) \tag{1}$$

where $t \in \{0, 1, \ldots, T\}$ is a timestep, $T = 1000$, $\bar{\alpha}_t = \prod_{k=1}^{t} \alpha_k$ and α_i are pre-defined according to a noise schedule. In a reverse process, the noise latent z_T is used as the seed for a denoising chain, based on the network ϵ_θ, which aims to recover the noiseless latent z_0. This is finally fed to a decoder $\mathcal{D}$ to recover the image. The network parameters θ are trained to predict the noise ϵ_t introduced at each step of the forward process, using the loss

$$L_{SD} = \mathbb{E}_{z, \epsilon_t \sim N(0,I), t} \|\epsilon_t - \epsilon_\theta(z_t, t, c)\|_2^2. \tag{2}$$

During inference, the reverse process generates a clean latent $\hat{z}_0$, starting from a random seed $z_T \sim N(0, 1)$, conditioned on a text embedding c. For stable diffusion, the model ϵ_θ has a UNet architecture with 4 downsample, 1 mid, and 4 upsample blocks, as shown in Fig. 2.

3.2. Style attribution

Style attribution aims to identify in a reference dataset $\mathcal{D} = \{x_1, \ldots, x_n\}$ of images x_i, usually but not necessarily the dataset used to train the diffusion model, the subset of images of style closest to that of a query image q, which is usually an image synthesized by the model. Like all retrieval operations, this requires two components: 1) a feature representation $\mathcal{F} : x \rightarrow f$ that maps each image x into a feature tensor f that captures the stylistic elements of x, and 2) a similarity metric $d(\mathcal{F}(q), \mathcal{F}(x))$ that quantifies the style similarity of the feature tensors extracted from query q and reference x images. This allows retrieving the images in $\mathcal{D}$ that are most similar to q, which can then be inspected, e.g., to determine copyright violations.

3.3. Introspective style attribution

Style attribution is not a widely studied problem in the literature. Previous approaches [38, 43] attempted to train an external feature representation $\mathcal{F}$ or adapt existing feature representations, such as DINO [27], by addition of parameters specific to the extraction of style features. This has two major drawbacks: it requires computationally intensive large-scale training to cover the diffusion model's sampling distribution, and the external feature extractor $\mathcal{F}$ introduces substantial memory and computational overhead—often exceeding the diffusion model's resource requirements.

Introspective style attribution aims to solve the problem without external models by using only the features computed by the denoising network ϵ_θ of the diffusion model. This is inspired by recent showings that this network produces feature representations that 1) disentangle image properties such as structure and color [42, 49] and 2) are sufficiently informative to solve classical vision problems such as classification [45], correspondence [40], or segmentation [41] with good performance. This suggests that these features should also be sufficient for style attribution. In what follows, we show that this is indeed the case and can be achieved with very simple feature representations and popular similarity metrics, resulting in the simple `IntroStyle` approach, based on the operations of Fig. 2. The features and similarity metric used by `IntroStyle` are discussed next.

`IntroStyle` Features To compute a feature representation, image I is first mapped into a latent vector $z_0 = \mathcal{E}(I)$, which is noised to a time t using Eq. 1. The noised latent z_t is then passed through the denoising network ϵ_θ with null text embedding $c = \varnothing$. A feature tensor F^t is then extracted from intermediate network layers. Given the UNet architecture commonly used to implement ϵ_θ, we consider this model an autoencoder that stacks a sequence of feature encoder and decoder blocks. The encoder/decoder split is quantified by equating all layers up to the upsampling block idx as an encoder and the remaining upsampling layers as a decoder. The feature tensor $F^{t,idx}$ is finally extracted from block idx, as illustrated in Fig. 2 for $idx = 1$.

To derive a compact feature representation that captures style information, we then take inspiration from the literature on style-aware generative models and the successful StyleGAN approach to control style by Adaptive Instance Normalization (AdaIN) of first and second-order feature statistics [12]. This leads us to hypothesize that diffusion models may be intrinsically producing similarly normalized

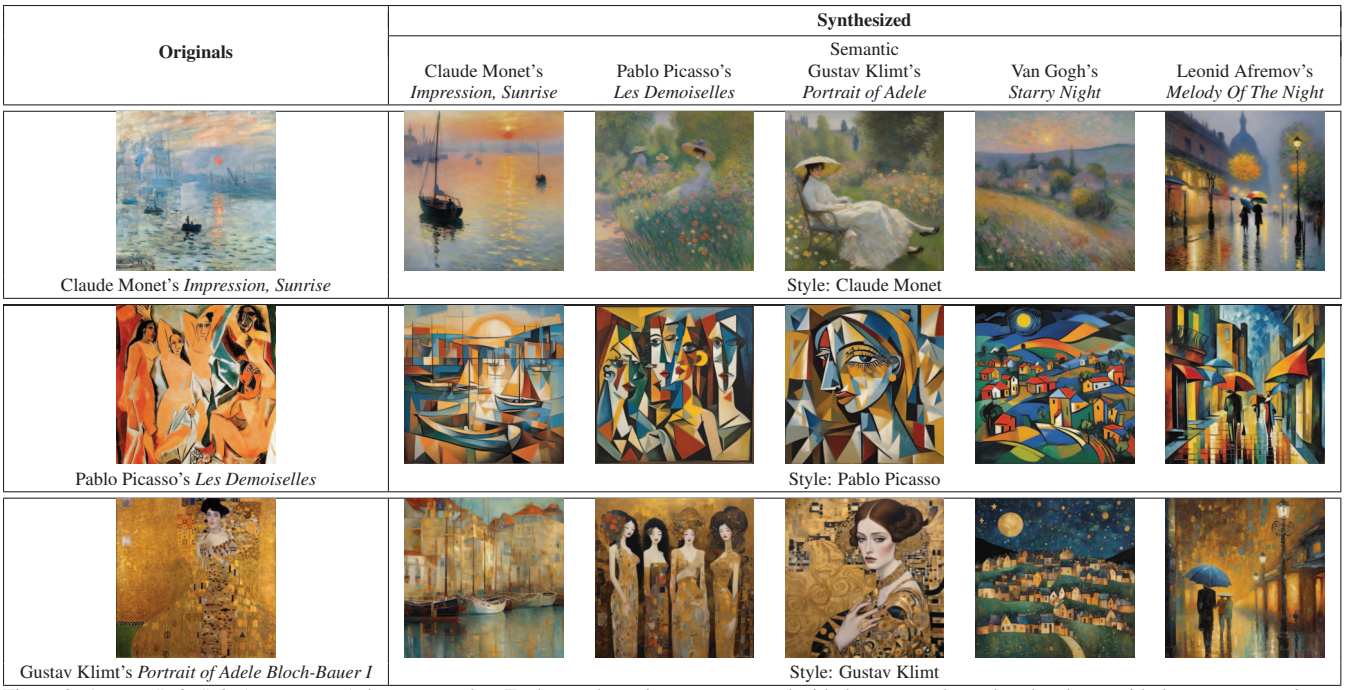

Figure 3. *Artistic Style Split* (`ArtSplit`) dataset samples. Each row shows images generated with the same style, and each column with the same semantics.

features, in which case these statistics should be highly informative of style and eliminate dependence on pixel-wise correlations.

Based on this intuition, we propose a simple style feature representation for `IntroStyle`, based on the channel-wise mean and variance of feature responses. Given the feature tensor $F^{t,idx}$, the mean and variance of channel c are obtained with

$$\mu_c = \frac{1}{WH}\sum_{i=1}^{W}\sum_{j=1}^{H} F^{t,idx}_{c,i,j}\,,\ \sigma_c^2 = \frac{1}{WH}\sum_{i=1}^{W}\sum_{j=1}^{H}(F^{t,idx}_{c,i,j}-\mu_c)^2$$

where W, H are the spatial dimensions of the tensor. The style feature representation of the image I is finally

$$f^{t,idx}(I) = (\mu_1, \ldots, \mu_C, \sigma_1^2, \ldots, \sigma_C^2)^T \tag{3}$$

where C is the number of channels at block idx. Both t and idx are hyperparameters of the method, which is summarized in Fig. 2.

Similarity Metric The `IntroStyle` representation of an image models its feature statistics with a multivariate Gaussian distribution of C dimensions. As shown in Fig. 2, Gaussian parameters (μ_1, Σ_1) and (μ_2, Σ_2) are extracted for images I_1 and I_2, respectively. Hence, any similarity measure between two multivariate Gaussians can be used to compare the images. We have investigated several similarity measures and found the 2-Wasserstein (W_2) distance

$$W_2((\mu_1,\Sigma_1);(\mu_2,\Sigma_2))^2 = \|\mu_1-\mu_2\|_2^2 + \operatorname{tr}\left(\Sigma_1+\Sigma_2-2(\Sigma_1^{1/2}\Sigma_2\Sigma_1^{1/2})^{1/2}\right). \tag{4}$$

to be effective for style retrieval. A comparison to other metrics, including Euclidean (L_2) Distance, Gram Matrix, and Jensen-Shannon Divergence (JSD), is presented in the experiments. In all cases, the covariance matrices are assumed diagonal with the variances of the `IntroStyle` representation as diagonal entries.

4. The `ArtSplit` Dataset

To address the limitations of existing datasets for fine-grained evaluation of style retrieval, we propose the *Artistic Style Split* (`ArtSplit`) dataset. This was created with the prompt-image pairs of the 2 most recognized works of 50 prominent artists from the LAION Aesthetic Dataset [35]. For each of the two paintings, ChatGPT[23] was asked to generate a "style" specification and a "semantic" description, such that there is no style information in the semantic description and vice-versa. Stable Diffusion v2.1 was then used with a combination of two prompts, "style" and "semantic," to synthesize a reference image dataset. With 50 artists ("styles") and 100 paintings ("semantics"), this led to $50 \times 100 = 5,000$ prompt combinations. A set of 12 images was sampled per combination, yielding $60,000$ images in total. The procedure is detailed in Supplemental Section 4.

Fig. 3 shows some of the images in the dataset. The left column shows three of the 100 original paintings used to produce the style and semantic descriptions, e.g. style: "Gustav Klimt" and semantic: "Portrait of Adele". The remaining table entries are images synthesized for different combinations of style (row) and semantic (column). Note

(a) Queries (b) `IntroStyle` (ours) (c) CSD [38] (d) GDA [43]

Figure 4. Image retrieval on WikiArt Dataset for `IntroStyle`, CSD, and GDA. The first column (left to right) shows the query image, followed by `IntroStyle`'s top 3 retrieval results, CSD and GDA, respectively. Green colors indicate correct and red for incorrect retrievals.

	WikiArt						DomainNet					
	mAP@k			Recall@k			mAP@k			Recall@k		
Method	1	10	100	1	10	100	1	10	100	1	10	100
VGG-Net Gram [8]	0.259	0.194	0.114	0.259	0.527	0.804	-	-	-	-	-	-
SSCD ResNet-50 [26]	0.332	0.243	0.135	0.332	0.591	0.839	0.713	0.672	0.608	0.713	0.964	0.998
MoCo ViT-B/16 [10]	0.440	0.332	0.188	0.440	0.689	0.879	0.784	0.757	0.708	0.784	0.960	0.997
DINO ViT-B/16 [3]	0.458	0.348	0.197	0.458	0.706	0.889	0.774	0.741	0.684	0.774	0.956	0.997
CLIP ViT-L [27]	0.595	0.489	0.316	0.595	0.830	0.952	0.928	0.923	0.778	0.928	0.972	0.996
GDA ViT-B [43]	0.439	0.331	0.186	0.439	0.685	0.873	0.800	0.768	0.694	0.800	0.966	0.998
GDA CLIP ViT-B [43]	0.522	0.419	0.259	0.522	0.782	0.935	0.817	0.786	0.716	0.817	0.972	0.998
GDA DINO ViT-B [43]	0.458	0.349	0.199	0.458	0.709	0.8888	0.770	0.739	0.684	0.770	0.956	0.997
CSD ViT-L [38]	0.646	0.538	0.357	0.646	0.857	0.956	0.833	0.811	0.766	0.833	0.972	0.998
`IntroStyle` (Ours)	0.887	0.852	0.525	0.887	0.909	0.941	0.954	0.949	0.850	0.954	0.982	0.995

Table 1. Evaluation on WikiArt and DomainNet datasets. `IntroStyle` outperforms all previous methods significantly. The best values are highlighted with red and second best with orange.

how the procedure creates a fine-grained sample of the style $\times$ semantic space.

Style attribution performance is measured using the original artist's paintings (left column at Fig. 3) to retrieve images. Each original is used as a query, the dataset images are ranked by similarity to this query, and performance is evaluated by the agreement between the labels of the retrieved images and that of the query. This is measured using a weighted score of Top-k retrieved neighbors that are correctly matched. Based on the labels, we perform two assessments on the retrieved images. Style labels allow the evaluation of style attribution accuracy. On the other hand, semantic labels allow the evaluation of semantic similarity and reveal the model's semantic bias. A good representation for style attribution should simultaneously achieve high style and low semantic retrieval accuracy.

5. Experiments

We use the Stable Diffusion v2.1 model initialized with weights from Hugging Face. Introspective attribution is training-free; we extract feature tensors from the model to obtain the `IntroStyle` image representation using an NVIDIA RTX 3090ti GPU with batch size 4. Each image is center-cropped to resolution 512×512. The guidance scale is irrelevant since we do not use a text prompt. Unless otherwise noted, `IntroStyle` is implemented with $idx = 1$, timestep $t = 25$, and the 2-Wasserstein distance.

5.1. Datasets and Baselines

Dataset. *DomainNet* [24] comprises images from six domains: Clipart, Infograph, Painting, Quickdraw, Real, and Sketch, with approximately equal representation. Due to strong stylistic similarities between Quickdraw and Sketch domains, we excluded the former. The test set, consisting of $127,855$ images, was randomly divided into $25,571$ query images and $102,284$ reference images. *WikiArt* [34] contains $80,096$ fine art images representing $1,119$ artists across 27 genres. We randomly split this dataset into $64,090$ reference and $16,006$ query images. As proposed in [38], each artist defines a separate class, correlating style and artist. WikiArt poses a greater style retrieval challenge since the large number of artists makes chance level matching performance 0.09%, significantly less than the 20% of DomainNet. These are the only two publicly available datasets for style retrieval[1], making the proposed `ArtSplit` dataset a valuable addition to the field. We also conduct experiments on `ArtSplit`, which has 100 queries, each ranking $60,000$ images.

Evaluation Metrics. Style attribution performance is measured by two standard retrieval metrics [38, 43]: Recall@k and mean Average Precision (mAP)@k, for $k \in 1, 10, 100$.

Baselines. We compared `IntroStyle` to several baselines from the style attribution and representation learn-

[1]The Behance Artistic Media (BAM) dataset [32] is not publicly available.

(a) Queries (b) `IntroStyle` (ours) (c) CSD [38] (d) CLIP [27]

Figure 5. Image retrieval for style-based evaluation on `ArtSplit` Dataset, with images ranked highest to lowest from left to right. We show the ranked images for a fixed semantic for isolating stylistic variations. Green colors indicate correct and red for incorrect retrievals.

(a) Queries (b) `IntroStyle` (ours) (c) CSD [38] (d) CLIP [27]

Figure 6. Image retrieval for semantic-based evaluation on `ArtSplit` Dataset, with images ranked highest to lowest from left to right. The results suggest that our retrieval emphasizes styles rather than semantic content. Green colors indicate correct and red for incorrect retrievals.

Method	Style-Eval@10 ↑	Semantic-Eval@10 ↓
DINO ViT-B/16 [3]	0.682	0.190
CLIP ViT-L [27]	0.642	0.223
GDA DINO ViT-B [43]	0.541	0.143
CSD ViT-L [38]	0.636	0.183
`IntroStyle` (Ours)	0.823	0.114

Table 2. Evaluation results on the `ArtSplit` dataset.

ing literature. For style attribution, we consider two recent methods, Contrastive Style Descriptors (CSD) [38] and Generative Data Attribution (GDA) [43][2], which employ fine-tuning on additional datasets for enhanced performance. This is complemented by a training-free method based on VGG Gram Matrices [8]. We utilize the final layer embeddings as feature representations for all models under comparison. This standardized approach ensures a fair comparison across diverse architectures and training paradigms, allowing us to isolate and evaluate the effectiveness of each representation for style attribution. For representation learning, we consider multiple vision foundational models, including CLIP [27], DINO [3], MoCo [10], and SSCD [26], which have proven effective in the literature for various computer vision tasks.

5.2. WikiArt and DomainNet Results

Table 1 shows that `IntroStyle` significantly outperforms the various baselines for style attribution the WikiArt and DomainNet datasets. Its gains are quite significant for the more practically relevant (lower) values of k. In the more challenging WikiArt dataset, `IntroStyle` achieves a gain of close to 20 points over the previous best approach (CSD) for both mAP and Recall when $k = 1$. For $k = 10$, the mAP gain is more than 30 points. The improvement drops at $k = 100$ because most artists have less than 100 images, making it impossible to achieve a perfect mAP score and masking the differences between methods. In DomainNet, when compared to the baselines, `IntroStyle` has gains of around 10% for $k = 1$ in both the metrics and a similar improvement on mAP@10. We note that these are gains of large magnitude, not commonly seen in retrieval applications, suggesting that introspective style attribution is vastly superior to methods based on external models. Fig. 4 illustrates this by showing examples of the three top matches retrieved by `IntroStyle`, CSD, and GDA for the queries on the left. The middle row shows how the baselines are prone to retrieving images semantically similar to the query rather than of the same style, while `IntroStyle` is more focused on style.

5.3. `ArtSplit` Dataset Results

For the style-based evaluation, we calculate the rank of the 12×100 positives, averaged for the 100 different query images, and semantic-based evaluation is done for 12×50 positives, averaged for the 100 queries. The higher the score for style-based assessment, the better, and the lower, the better for semantic-based. If a model is truly agnostic to semantics and solely focuses on style, the semantic-based retrieval should be random and around 0.01.

Table 2 compares different approaches on the `ArtSplit` dataset. In particular, `IntroStyle` outperforms all the models in both evaluations by 20% in style and 8% in semantics compared to the second-best method, showing its effectiveness at isolating style. Fig. 5 shows

[2]We use the `real mapper` to obtain features

Block	mAP@k			Recall@k		
	1	10	100	1	10	100
$idx = 0$	0.866	0.830	0.474	0.866	0.879	0.911
$idx = 1$	0.887	0.852	0.525	0.887	0.909	0.941
$idx = 2$	0.888	0.852	0.521	0.888	0.907	0.938
$idx = 3$	0.872	0.836	0.491	0.872	0.886	0.916

Table 3. Ablation on feature block index selection, all with $t = 25$.

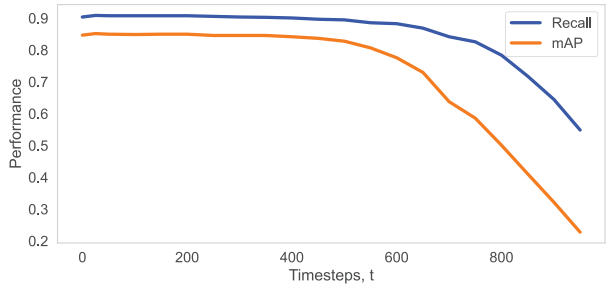

Figure 7. Effects of different timesteps on WikiArt image retrieval.

some results of the style-based evaluation, illustrating how `IntroStyle` has a significantly higher capacity than the baseline representation to focus on stylistic features. Our method ignores semantic details, as confirmed in Fig. 6, which refers to the semantics-based evaluation and shows that existing methods have a much stronger tendency to retrieve images of semantics similar to the query.

5.4. Model Size

Table 5 compares model sizes across state-of-the-art methods. Our approach (`IntroStyle`) has the highest parameter count primarily due to the large Stable Diffusion UNet, which represents a limitation of our method as it requires more substantial computational resources.

5.5. Ablations

In this section, we ablate various parameters and discuss the best choices for timestep t, block index idx, and similarity metric. All experiments are performed on WikiArt.

Timestep. Fig. 7 shows how style attribution performance (Recall and mAP at $k = 10$) varies with the timestep $t \in [0, 950]$ for the implementation of `IntroStyle` with features of up block index $idx = 1$ and the 2-Wasserstein distance. The performance remains stable until around $t = 400$, dropping significantly after that. We choose $t = 25$ as the default value for the timestep hyperparameter of `IntroStyle` as this performed the best on WikiArt and DomainNet. Similar observations hold for up block indices $idx \in 0, 2, 3$, which are presented in the supplementary.

Up block index. Table 3 compares the performance of `IntroStyle` with $t = 25$ and the 2-Wasserstein distance for different values of the up block index idx. The best overall performance on both datasets is achieved when $idx = 1$, but `IntroStyle` is robust to the choice of features used to create the feature representation. We compare the model size for each idx in supplementary.

Metric	mAP@k			Recall@k		
	1	10	100	1	10	100
L_2	0.888	0.852	0.521	0.888	0.907	0.938
Gram	0.869	0.835	0.491	0.869	0.885	0.916
JSD	0.888	0.849	0.525	0.888	0.908	0.940
W_2	0.887	0.852	0.525	0.887	0.909	0.941

Table 4. Comparison on different metrics.

Method	Number of Parameters
DINO ViT-B/16 [3]	21M
CLIP ViT-L [27]	427M
GDA DINO ViT-B [43]	86M
CSD ViT-L [38]	305M
`IntroStyle` (idx = 1)	808M

Table 5. Comparison on the model sizes.

Similarity metric. Table 4 compares the performance of `IntroStyle` for the different similarity metrics of Section 3.3, on WikiArt, using $t = 25$ and $idx = 1$. The 2-Wasserstein distance has the best overall performance, albeit it is only marginally better than the Euclidean Distance and the JSD. This shows that `IntroStyle` has a fairly stable performance with respect to the choice of similarity measure. We choose the 2-Wasserstein distance as the default metric for implementing `IntroStyle`.

6. Discussion

We presented `IntroStyle`, a training-free style attribution model that extracts stylistic features to compute similarity scores between images. `IntroStyle` was shown to outperform existing attribution methods across all metrics. To address the lack of datasets to evaluate style attribution, we developed `ArtSplit`, a synthetic dataset that isolates stylistic elements and enables fine-grained attribution evaluation. The robust performance of `IntroStyle` on both this and previously used datasets establishes it as a state-of-the-art method for style attribution, with potential applications in protecting artists' intellectual property rights.

`IntroStyle` has 2.5× more parameters than CSD (Table 5), which constrains its deployment on memory-limited devices. However, it can leverage the features produced by a diffusion model, without additional memory overhead, for style attribution of the images generated by that model. While our current evaluation focuses on Stable Diffusion v2.1, future work could explore compatibility with other pre-trained diffusion models, investigate performance gains through additional linear/MLP layers, and examine applications in diffusion-based style transfer.

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